

B.Tech. Degree IV Semester Examination April 2014**IT 1404 DATABASE MANAGEMENT SYSTEMS**
(2012 Scheme)

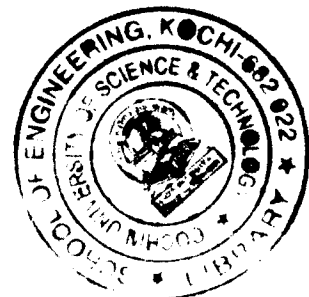
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) Describe the need for database schema. Draw an example schema. Also compare it with database state.
- (b) Define weak entity type. Explain its features and draw the symbol used in E R diagram.
- (c) Distinguish between heap file and sorted files. How can you delete a record from the file?
- (d) Discuss any two techniques for placing records on disk.
- (e) Write down Armstrong's inference rules. What do you mean by saying that "These rules are complete and sound"?
- (f) What is the need of normalization? Explain first three normal forms.
- (g) Explain schedule in transaction management.
- (h) Give features of data mining.

**PART B**

(4 x 15 = 60)

- II. (a) Construct an E-R diagram for a banking scenario. Assume in a city there are multiple banks and each bank has many branches, each branch has multiple customers and customers have various types of accounts. Some customers also had taken different types of loans from these bank branches, one customer can have multiple accounts and loans. (10)
 - (b) Discuss the main features of DBMS. (5)
- OR**
- III. (a) Draw the three tier architecture of a database management system and explain the function of each block. (8)
 - (b) List the main functions of database administrator and explain. (7)
- IV. (a) Describe seek time, latency and block transfer time. (5)
 - (b) Discuss the file organization method that takes less access time. Also explain the search algorithm. (10)
- OR**
- V. (a) Compare hashing and indexing (5)
 - (b) Explain dynamic hashing technique. Also write it's advantages. (10)

(P.T.O.)

- VI. (a) Write the relational algebra queries for the following: (9)
- Loan (branch-name, loan-no, amount)
Borrower (customer-name, loan-number)
Account (account-no, branch-name, balance)
Depositor (customer-name, account-no)
- (i) Find all loan numbers for loans with an amount greater than \$ 1500.
- (ii) Find names of all customers who have loan from the "Perryridge" branch and find the loan amount.
- (b) Define data independence. Explain the logical and physical data independence. (6)
- OR**
- VII. Define normalization. Explain 1NF, 2NF, 3NF and BCNF with examples. (15)
- VIII. (a) Describe the steps in building a datawarehouse (10)
- (b) List ACID properties of Transaction. (5)
- OR**
- IX. (a) Illustrate with an example an algorithm for testing conflict serializability of a schedule. (10)
- (b) Differentiate between serial and non serial schedule. (5)
